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Profits Integrity in the Motion Picture Industry Using Benford's Law: A Case Study of Toei Animation Company (2014–2024)

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Abstract ;

This study aimed to detect profits integrity of Toei Animation Company during the period (2014–2024) using Benford's Law. Toei Animation, the leading company in the anime films industry was selected for analysis. The sample included 176 observations across 16 accounting items. The analysis indicated stable profit growth and strong consistency between gross and net profits. Results from the Chi-Square test and the mean absolute deviation (MAD) metric demonstrated that the data conformed to Benford's Law, supporting the integrity of the company's profits. The findings further suggest that Toei Animation Company exemplifies accounting transparency and fairness within the motion picture industry.

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1. *Introduction*

Profit integrity is one of the main concepts in contemporary financial accounting. It reflects the honesty and reliability of the reported accounting information and their ability to represent the company’s actual performance. Profit integrity also provide a fair view of the financial statements disclosed to the company’s owners and stakeholders, thereby enhancing the confidence of investors and auditors in the reported information. With globalization, technology and artificial intelligence, motion picture or film industry has become more sophisticated, prompting film companies to refine their business models to remain in the market (Vitkauskaitė, 2020, p2). On the other hand, accounting function in motion picture companies considered one of the most important contemporary issues, it aims to provide statements that enable the assessment of a company's financial position, such as the balance sheet and its profitability, i.e., the income statement. Information provided by financial statements is often the only source related to the company's view, therefore, some motion picture companies attempt to present their financial position closer to actually needs, whether in terms of their performance or financial position by using margins left by some accounting rules, and even loopholes or accounting flexibility (Bonnet, 1995, p5). Accounting is also an information system and one of the most important source of organized information, so, if accounting provides information about a specific company's activity, it is important to accurately understand this information and determine its quality and integrity (Azevedo, Vieira, Marques, & Almeida, 2024, pp76-77).

Digital technology has revolutionized all stages of motion picture production, from scriptwriting and filming to editing and production, and finally distribution and screening on digital platforms. Digital platforms such as Netflix, YouTube, and Amazon Prime offer today new opportunities to reach a huge audience without the need for large production costs (Abdelhak & Abd Alaal, 2025, pp13, 15).

Many authors use the term "creative accounting" for all practices of earnings management or income smoothing derived from established accounting standards and interpreted through sophisticated accounting construction techniques. A company can falsify its financial statements by overstating its revenues, under recording its expenses, and misrepresenting its assets and liabilities (Luty, 2022, p 21). Webster's New World Dictionary defines manipulation or fraud as a deliberate deception of a person into giving up his property or a legitimate right. Its legal definition includes: "A general term encompassing all different tools devised by human ingenuity, by which an individual gains an advantage over another through false suggestion and concealment of truth using surprise, deception, cunning and any unfair means by which others are deceived." It is an intentional act committed to harm or injure others for unfair or illegal gain (Rezaee & Riley, 2010, pp4-5). If we distinguish between creative accounting practices and accounting fraud, we must present the two concepts. Mulford & Comiskey defined creative accounting as all steps used to play the financial numbers game, including aggressive choices between accounting principles, within the boundaries of accounting principles, fraudulent financial reporting, and steps taken toward earnings management and income smoothing (Mulford & Comiskey, 2002, p15). On the other hand, accounting fraud involves the intentional distortion of the truth, or the use of deception to obtain unfair or illegal financial advantages, and intentional misrepresentation affecting financial statements by one or more individuals among management personnel or third parties (Jones, 2010, p6). So creative accounting practiced without violating the conceptual framework of accounting principles and standards, but rather using accounting flexibility. Meanwhile, accounting fraud is committed by violating these accounting principles and standard. Based on the above, it is essential to explore the most effective approaches to ensure the quality and integrity of reported profits in all sectors. The motion picture industry in the 1980s and early 1990s considered a major hotbed of creative accounting practices, primarily due to the special accounting system applied to this sector, as film production and post-production often carried out in different accounting years. The standard applied was Financial Accounting Standard (FAS 53: Financial Reporting by Producers and Distributors of Motion Picture Films) issued by the Financial Accounting Standards Board (FASB) (De La Torre, 2009, p106). On the other hand, many researchers have questioned about methods used by film production companies to distribute profits, and the integrity of its revenues. From this standpoint, profit integrity defined as a state or condition of completeness, impeccability, soundness, and completeness in the disclosure of earnings in a company's financial statements, thus being free from any intention to distort or manipulate accounting (Orlitzky & Monga, 2018, p13). According to many researchers, there are many creative accounting techniques practiced by motion pictures companies. We present some of these practices as follows:

1.1. Real earnings management (REM):

Through its study, George Foster examines and analyzes the real- practices of earnings management (REM) in the film industry, commenting primarily on the study by Gong, Young, and Zhou in 2023. He touched upon the technique called Real earnings management (REM). The researchers hypothesize that publicly listed film studios will have a greater incentive to accelerate the release date of films expected to generate high revenues, when facing pressure to increase quarterly revenues and profits (George Foster, 2023, p1251). In other words, when listed film production companies face the risk of declining quarterly revenues—especially when previous films' revenues are low—they accelerate the release of films expected to generate high revenues and include those revenues in the current quarter's results. This behavior is a classic example of real earnings management (REM), as it is not merely an accounting adjustment (as in earnings management), but rather an actual change in management's operational decision (advancing the film's release date) aimed to influence revenues reported in the current quarterly financial reports. This intended to avoid any negative reaction from the financial market or investors and improve the company's view. Here, the company is not practicing accounting fraud, but creative accounting. When it expects its quarterly profits to be weak due to films that do not perform well, it advances the release of strong films to improve revenues in that financial quarter.

Some researchers such as Rivera & Foderick explained the creative accounting practices in the film industry as a term refers to a set of opaque or creative accounting techniques used by the film, television, and music industries, which include (Rivera & Foderick, 2024, p1):

1.2. Cross-collateralizing losses from different films production to decrease or eliminate the net income owed to creators, investors, and other profit participants:

For example, a company can use losses from different films to reduce the net profits owed to profit participants such as directors, writers, actors, investors, or other parties sharing a percentage of the profits. This means that the studio combines the results of several films, so that the losses of one film counted against the profits of another, resulting in a decrease or disappearance of the apparent net profits. Consequently, profits will not paid to participants who contracted for a percentage of the "net profits."

1.3. Inflating expenses to decrease profits:

Films Expenses inflated to decrease profits through another technique named "Accrual-Based Earnings Management", this technique "Accrual accounting" means exploiting the time or discrepancy between the date of recording an accounting transaction and the date of actual receipt or payment of expenses or revenues, i.e. manipulation through the accrual basis stipulated in international accounting standards IAS-IFRS as a basic accounting assumption.

1.4. Charging fees from one subs-entity to another in order to reduce the distributed profits:

For example, the big film companies usually made up of several entities: a production entity, a distribution entity, and a marketing entity. Here, the producing entity pays a "distribution fee" to the distributing entity (they are parts of the same group). This means that artificial internal fees imposed between entities within the same group. Here, the company has paid amounts that do not go beyond its consolidated balance sheet, allowing profits shifted to areas where there are fewer financial taxes (or where taxes are lower). As a result, many commercially successful films that generate millions of dollars in revenue recorded as losses on the studios' financial statements. Numerous actors, writers, and producers who have recovered hundreds of millions of dollars in profits and damages have challenged this practice; many of them continue to fall prey to these widely fraudulent accounting practices in the world (Rivera & Foderick, 2024, p1). Profit-sharing claims can also filed when film production companies do not receive their fair share of profits according to contracts agreed upon before working on the film. (Strum, 2017, p 458).

Producers, supporting actors, directors, writers, co-producers, and sometimes investors often use Net Profit Participation Contracts but creative accounting practices allow the studio chain to show the film as a "loss," resulting in no profits being paid to them. Distributors are often the primary beneficiaries of these practices; this excludes well-known stars who play leading roles and have negotiating power. They can impose a percentage of the film's gross profit in the contract before deducting costs, or impose a high fee upfront.

This study aims to present the latest tools used to detect the integrity of reported profits by motion picture companies. We chose in this study the Japanese company Toei, given its prestigious position in the motion picture and animation production industry. The study sample consists of 176 observations representing 16 major accounting items, collected annually over the period 2014–2024. This research examine the extent to which the company's accounting data comply with Benford's Law by applying Chi-Square Test and Mean Absolute Deviation (MAD), in addition to conducting a consistency analysis of the main accounting items especially gross and net profits, in order to detect the degree of profit integrity in Toei Company during the period 2014–2024.

The main research question of this study formulated as follows:

To what extent do the reported profits of Toei Company during the period (2014–2024) reflect integrity in light of Benford's Law test?

Based on the main research question of the study, the study hypothesis formulated as follows:

The distribution of Toei Company's accounting data for the period 2014–2024 conforms to Benford's Law, thereby indicating the good integrity in the company's reported profits.

This study contributes to the literature on earnings quality and integrity in the motion picture industry. It distinguishes itself from previous research by providing empirical evidence on profits integrity of Toei Company, a Japanese company recognized for its transparency in financial reporting. The study employs both consistency analysis and Benford's Law as a modern statistical tool widely used by auditors to detect integrity and disclosure reliability.

The remainder of this paper organized as follows: Section 1 presents the Introduction and theoretical background.

Section 2 describes the data and methodology of the empirical study. Section 3 reports and discusses the results. Finally, we conclude with key implications and recommendations to enhance profits integrity in the motion picture industry.

2. Methodology

Toei Company is a motion picture company recognized for its sustainability and its high degree of transparency in financial reporting. This Company founded in 1948 as NIHON DOGA. LTD in Shinjuku-ku, Tokyo, Japan, to produce animation. In 1952, the company's name changed to NICHIDO EIGA (Film), than, NICHIDO FILM CO purchased by TOEI CO, and renamed TOEI DOGA CO; its headquarters located in Tokyo. In October 1998, the company's official name changed to Toei Animation Co., Ltd. Its main activity is the production, marketing and licensing animation in Japan and internationally. The company listed on the JASDAQ Stock Exchange on December 2000. The company aims to produce animation for TV and movie theaters, and to broadcast them across various media such as TV, movie theaters, DVDs and Blu-ray discs, streaming platforms, smartphones, and more. Animation production is the starting point for Toei Animation. It has produced a large number of works and has achieved significant successes since its founding. Today, it is active in publishing its works on streaming platforms, especially Blu-ray and DVD discs of its animation videos. The company is interested in producing, marketing and licensing the following: TV animations, Theatrical Features, video package (Blu-ray disc/DVD), video streaming, and online game applications

(Toei Animation, Ltd., n.d, https://corp.toei-anim.co.jp/en/company/outline.html).

The study sample consists of 176 observations, representing 16 key accounting items reported annually during the period 2014-2024 using the financial statements of Toei Company. The financial items examined to assess the potential for profit integrity extracted annually from the balance sheet and income statement of Toei Company during the period 2014-2024.

To test the study hypotheses, we applied Benford's Law distribution to detect the profit integrity reported in the company's financial statements. The analysis focused on the first digits of amounts of the selected items and comparing them with the expected Benford distribution.

Simon Newcomb discovered the phenomenon of Benford in 1881; he noticed that the first pages of logarithmic tables contained smaller, full numbers than the subsequent ones. His observations ignored and rediscovered by Frank Benford in 1938. Benford discovered that most of the first pages of a logarithms book looked impressive because they were sparse compared to the last book pages. He reached the same conclusion as Newcomb in previous years: that people often look for numbers that begin with low digits rather than high digits. He hypothesized that there are many numbers begin with low digits. Benford tested this hypothesis by studying a set of numbers that included baseball statistics, stock prices, city populations, newspapers, death rates, cost data, and other data. Experimental observations from tabular records confirmed that the probability of generating the first multiple numbers beginning with (1) was actually higher than numbers beginning with (9). The economist Varian was the first who suggest the possibility of using Benford's law as a test of the validity and significance of random scientific data within the social sciences in 1972. Benford's law has been widely used since the 1980s in the fields of statistics, economics, and accounting, as well as in engineering and physics. Benford's Law applied as a measure of the reliability of data, when data does not conform to Benford's Law, it indicates that some of the data preparation manipulated or otherwise distorted. From this perspective, Benford's Law is very important in assessing accounting manipulation. Therefore, Benford's Law allows auditors to focus their attention directly on hidden areas or locations of manipulation and detect anomalies in financial operations, thus uncover and assess accounting manipulation practices. The idea behind Benford's law is to use numerical analysis to detect anomalies in financial transactions; he made some assumptions regarding the geometric pattern of the applied phenomena and formulated the expected patterns of numbers from spreadsheets. The expected frequencies arranged in order of the first digit, as in the following equations:

The most famous equation in Benford's Law used to calculate the probability of a particular number appearing in the first place (first digit) of a number:

$$P(D_1 = d_1) = Log\ 10\ (1 + (1 / d_1))\ ;$$
$$d_1 = \{1, 2, 3, \ldots 9\} \qquad (1)$$
$$P(D_1 = 1) \approx 0.301 \rightarrow 30.1\%$$

Where:

P: represents the probability of the observations occurring.

D: represents the rank of the number.

d: represents the number.

The following equation calculates the probability of a particular number appearing in the second digit (second digit) of a number:

$$P(D_2 = d_2) = \sum_{d1=9}^{9} Log10$$
$$(1 + (1 / d_1 d_2));$$

If we want to know the probability that the second number is 3, the equation adds the probabilities for all the numbers: {13, 23, 33, ..., 93}(Abed, 2019, pp277-279).

$$d_2 = \{0, 1, 2, \ldots 9\}\ ;\ d_1 = 9$$

Benford discovered that the numbers 1 through 9 not evenly distributed, with some numbers appearing more frequently than others do. He formulated the probability of the numbers (1 through 9) appearing in each position of the number.

Benford indicated that the probability of the number (1) appearing in the first position is higher than the number (2), the number (2) is higher than the number (3), and so on gradually until the number (9), which appears much less frequently in the first position, this is illustrated in the Table1(Mustapha, 2019, pp142-143):

Table 1. Probability Distribution of the First Digit according to Benford's Law

First Digit (d_1)	Benford's Probability	Benford's Percentage (%)
1	0.301	30.1%
2	0.176	17.6%
3	0.125	12.5%
4	0.097	9.7%
5	0.079	7.9%
6	0.067	6.7%
7	0.058	5.8%
8	0.051	5.1%
9	0.046	4.6%

Source: adapted from Mustapha (2019, p.143)

The steps to apply Benford's Law to assess the integrity of reported profits are as follows:

-Collecting data from Toei's annual financial statements;

-Using Excel program to present the Total sales development curves according to the company's various business segments, as well as its net sales volume during the period 2014-2024, then analyzing and interpreting their development during the same period.

-An initial detection of profit integrity by analyzing the consistency between the development of the selling, general and administrative expenses value (SG&A), the gross profit, and the net profit of Toei Company during the period studied;

-Extracting the first digit of each financial item using Excel;

-Calculating the actual distribution of the first digits (from 1 to 9) for each accounting item;

-Performing a Chi-Square test using SPSS to determine whether the actual distribution of the first digits differs significantly from the distribution predicted by Benford's law (Mustapha, 2019, p157);

-Calculating the mean absolute deviation (MAD) used as a quantitative metric to measure the extent, to which the observed distribution deviates from Benford's law, where:

MAD ⩽ 0.006 → **Close Conformity** with Benford's Law;

MAD between 0.006 and 0.012 → **Acceptable Conformity**;

MAD between 0.012 and 0.015 → **Marginally Acceptable Conformity** (additional tests recommended);

MAD ⩾ 0.015 → **Nonconformity** with Benford's Law (Bowman, 2016, p28).

The mean absolute deviation (MAD) was calculated using Excel program as follows:

$$MAD = \sum_{i-1}^{n} \frac{|xi - \bar{x}|}{n}$$

Where:

xi: Observed proportion of each first digit;

$\bar{x}$: Expected proportion from Benford’s distribution;

n: Number of categories, which equals 9 for the first-digit test (digits 1–9);

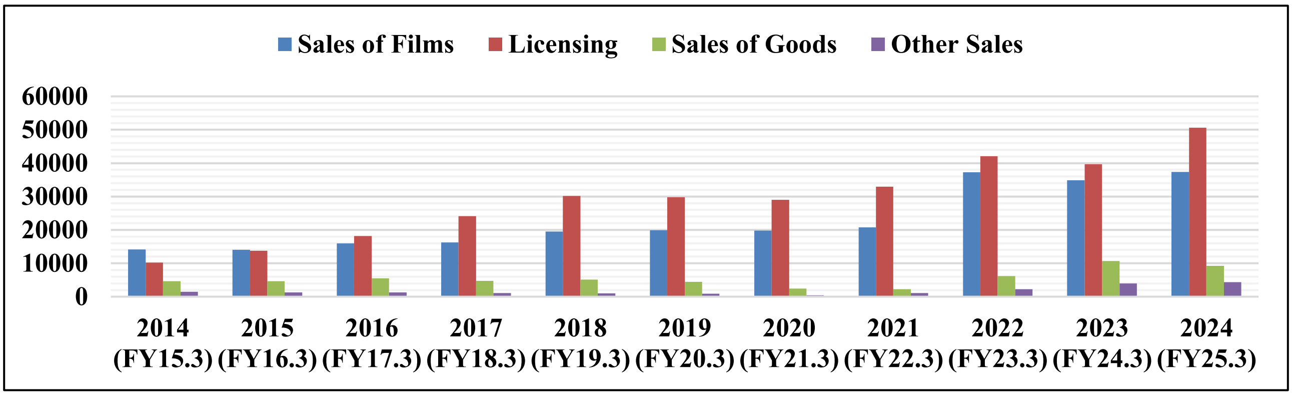
i: the index of each first digit category (Mean Absolute Deviation Formula, n.d, https://byjus.com/mean-absolute-deviation-formula/).

3. Results

In this section, we present the empirical results of the study and provide a discussion of their implications based on the Appendix Table A1.

The Figure 1 present the development of the company's sales across its main business segments: Film, Licensing, Sales of Goods, and others sales, during the period 2014-2024:

Figure 1. Toei Company's sales across its main business segments during the period 2014-2024



Source: Excel Outputs based on the Appendix: Table A1

For the films sector, the figure shows that it has increased from 14155 million yen in 2014 to 37323 million yen in 2024, a cumulative increase of 164%. The films sales gradually increased until 2021, then witnessed a strong jump after 2022, driven by successful film releases and international box office revenues. The licensing sector includes the licensing of intellectual property (anime and characters) for use in products, games, shows, and more. It considered the leader in revenue volume, having jumped from 10250 million yen in 2014 to 50582 million yen in 2024 with a cumulative increase of 393%. This reflects the company's expansion in exploiting its intellectual property (anime series and films, games, and many related products) in global markets. Since 2016, licensing revenue has surpassed film revenue and has become the company's primary revenue driver. For the Sales of Goods segment, Toei Company recorded revenues of 4628 million yen in 2014, which gradually declined during 2021 to 2231 million yen. The decline in 2020 and 2021 was due to the impact of the COVID-19 pandemic. However, a significant recovery began in 2022, achieving sales of 9211 million yen in 2024 due to the sale of new products, particularly goods related to the most popular anime characters, in addition to promotional goods through online stores.

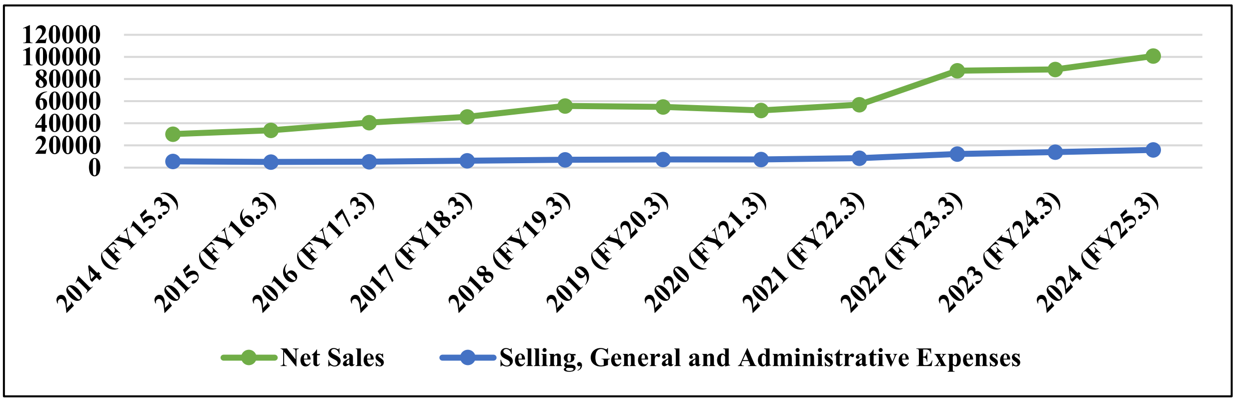
Other sales achieved relatively acceptable revenues, from 1437 million yen in 2014 to 4315 million yen in 2024. This includes sales of anime character stores and other online stores, as well as events such as live performances and musicals.

From the above analysis, we note that Toei Animation's sales witnessed strong growth during the period 2014–2024, led primarily by the licensing sector, which became the largest source of revenue. Films sector came in the second place, achieving significant growth, influenced by production and release cycles, followed by the other goods and sales of the company. It is clear that the company's strategic focus directed towards intellectual property globally through licensing, goods, and events.

As a general assessment of the company's products, we said that Toei Company witnessed a wide diversification during the period 2014-2024; its products cover anime films and series, licensing, goods, and events, which gives the company multiple sources of income and global appeal thanks to its internationally popular series and films, which continuously increases demand for its various products. Its successful strategy of achieving benefit and innovation in promotion by integrating physical stores, online stores, and live shows creates a comprehensive experience for fans.

Based on Figure 2, which presents the development curve of Toei company's Net Sales during the period 2014-2024 as follows:

Figure 2. Toei Company’s Net sales and operating expenses during the period 2014-2024



Source: Excel Outputs based on the Appendix: Table A1

The curves above present an upward trend over the period (2014-2024) with net sales rising from 30313 million yen in 2014 to 100836 million yen in 2024 as a peak value throughout the period.

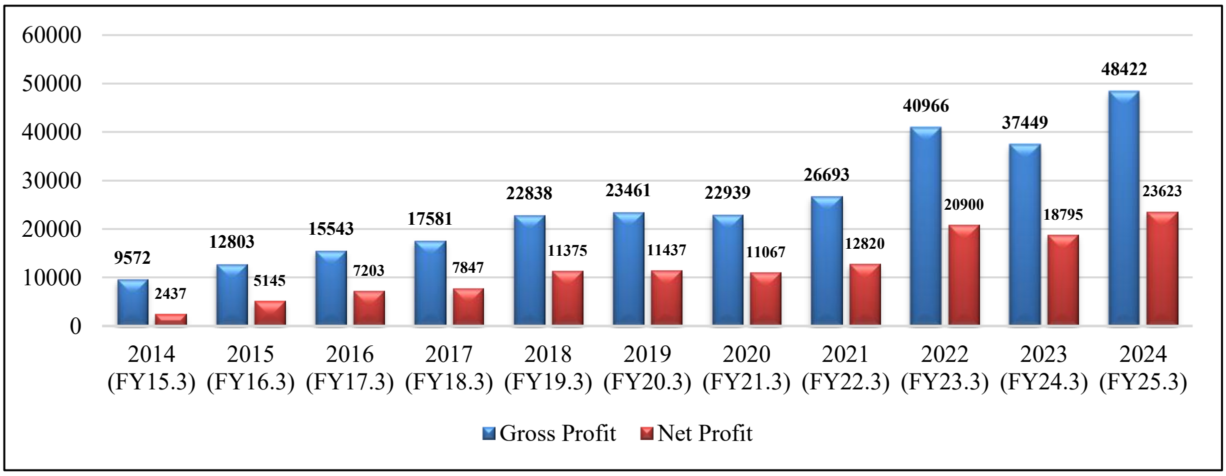
Steady, gradual growth in net sales, thanks to the global spread and growth of anime market and the expansion of digital platforms characterized the period (2014-2018).

The year 2019 also witnessed a significant improvement, thanks to the success of the company's major anime films. Net sales fluctuated during the (2020-2021) period, recording approximately 51595 million yen in 2020, and rose relatively to 57020 million yen in 2021. This fluctuation was due to the repercussions of the COVID19 pandemic, which affected film production globally through theater closures and the postponement of some releases, despite the partial compensation through the company's digital content sales. The period (2022-2023) recorded a significant jump in net sales, reaching 88654 million yen in 2023 but the year 2024 recorded its highest value during the period, thanks to the increase driven by anime productions, the strong resumption of film production, the return of events, and the expansion of global licensing sales in conjunction with the return of economic activity. In addition to the rising global demand for the company's anime as a cultural and entertainment product, and the continued expansion of international broadcasting. As for the company's operating expenses represented by selling expenses (marketing, advertising, and distribution of anime films and series), general expenses (communication, salaries, insurance,…etc), and administrative expenses, excluding the direct production costs of films or series. The figure 3 indicate the general trend of operating expenses compared to the company's net sales, which witnessed a relatively stable increase from 5576 million yen in 2014 to 8585 million yen in 2021, then increased significantly starting from 2022 to reach 15989 million yen in 2024. This is due to the expansion in spending on production and marketing activities in order to recover from the repercussions of the Covid19 pandemic. Compared to the company's net sales curve, we record a large gap in growth between net sales and SG & administrative expenses during the period 2014-2024, indicating that expenses are not at a level that may swallow revenues.

Therefore, it is noticeable that there are no direct signs of the company's expenses increasing, as net sales growth has often been faster or at least parallel to the expenses growth. We note also that expenses increased after 2022 slightly higher than the previous years, for the same reason mentioned above, related to the increase in operating expenses in the post-Covid19 pandemic period.

Figure 3 illustrates the company's gross profit and net profit curves over the period studied:

Figure 3. Toei Company's Gross and Net profit during the period 2014-2024



Source: Excel Outputs based on the Appendix: Table A1.

Gross Profit indicate the company's profit after deducting direct production costs (Cost of Goods Sold), reflecting the efficiency of production and pricing processes. However, net Profit is the final profit after deducting all operating expenses (such as SG&A), interest, taxes, etc. If gross profit and net profit increase at similar rates, this indicates the good control over expenses and it is a positive indicator of profits integrity. If a large gap appears (for example, a significant increase in Gross Profit but Net Profit does not follow), this may indicate an unexplained increase in expenses or the inclusion of abnormal items that affect the net results.

According to the Data of Toei Company during the period 2014-2024, we recorded the following observations:

-The Gross Profit and Net Profit curves move in the same direction over the period 2014–2024.

-There is no intersection or inversion between the two lines, meaning that the mathematical relationship between them is logical;

-The distance between the curves (the difference between gross and net) is relatively constant as business expands, but it has widened slightly in recent years as sales revenues have increased. This is normal because expenses rise with increasing business volume;

-There are no unjustified jumps or drops in net profit compared to gross profit;

-Based on the Appendix Table A1, the net-to-gross profit ratio, moves in a relatively stable range between 25% and 51%, with gradual increases during periods of significant profit growth (such as 2022, 2023, and 2024);

-There are no sharp jumps or unjustified fluctuations, reinforcing the impression that net profits are not inflated or understated relative to gross profits;

-Periods of rapid gross profit growth consistently reflected in net profits;

-The general pattern indicates a consistent relationship between gross profits and net profits, and no inflation in expenses.

According to the Figure 4, we tested the study's hypothesis. The results of Chi-square test revealed the following:

Figure 4. Chi-Square test during the period 2014-2024

Test Statistics

	Observed
Chi-Square [a]	,778
df	7
Asymp. Sig.	,998

Source: SPSS Outputs

We noted the following remarks:

-Calculated value: $\chi^2 = 0.778$;

-Degrees of freedom: df = 7;

-P-value (Asymp. Sig.)$\approx$ 0.998.

Since the significance value (p-value =0,998) is far greater than the 0.05, the study hypothesis is accepted. This indicates that the distribution of Toei Company's accounting data for the period 2014–2024 conforms to Benford's Law distribution, thereby indicating the good integrity in the company's reported profits. On the other hand, to confirm the previous result, we calculated the Mean Absolute Deviation (MAD) as shown in the Table 2, as a complementary measure to assess the conformity with Benford's Law:

Table 2. MAD calculation using the observed and expected Benford 'distribution

First Digit	Observed	Opserved (%)	Expected Benford's distribution	Expected Benford's distribution (%)	Absolute Diff (%)
1	57	32.386	52.976	30.100	0.023
2	30	17.045	30.976	17.600	0.006
3	27	15.341	22.000	12.500	0.028
4	17	9.659	17.072	9.700	0.000
5	13	7.386	13.904	7.900	0.005
6	8	4.545	11.792	6.700	0.022
7	10	5.682	10.208	5.800	0.001
8	7	3.977	8.976	5.100	0.011
9	7	3.977	8.096	4.600	0.006
Total	176	100	~176	100	**MAD = 0.011**

Source: Excel Outputs.

The MAD value (0.011) falls within [0.006 – 0.012], indicating an acceptable conformity with Benford's Law, thereby reinforcing the integrity of Toei Company's profits during the period 2014–2024.

4. Conclusion

Through this study, we detected the integrity of profits in the Japanese Toei Animation Company during the period 2014–2024, one of the most prominent motion picture companies in the world. The company known for its famous reputation in sustainability and its high degree of transparency in financial reporting. The study concluded a set of results, presented as follows:

-Toei Animation Company is one of the famous Japanese animation companies; it is characterized by its sustainability and achieving high annual profits;

-Toei Company characterized by the diversity of its activities and its stable range of sales, with a consistent relationship between gross profits and net profits.

-Based on the consistency analysis, we found that there is a good harmony between gross and net profits, which supports the integrity of profits in the period studied;

-The statistical tests conducted for the period 2014–2024 revealed a high degree of consistency between gross and net profits, along with a strong conformity to Benford's Law distribution.

-The Chi-square test results indicated that most of the examined accounting items did not show significant deviations from the expected Benford's distribution. This finding supports the integrity of the accounting items under study and, consequently, affirms the integrity of the company's reported profits;

-The results of the Mean Absolute Deviation (MAD) test also indicated that the good level of conformity with the Benford's distribution, thereby providing further support for the integrity of Toei Company's reported profits during the study period.

-Based on the previous results, we accept the study hypothesis and confirm that the observed distribution conforms to the expected Benford's distribution, thereby reflecting the good integrity of Toei Company's reported profits during the period 2014-2024.

Accordingly, the recommendations of our study formulated as follows:

-Toei Animation Company should maintain the integrity of its profits and sustain its excellence in the motion picture industry as a model for other motion picture companies to follow;

-Auditors are encouraged to expand the use of non-traditional statistical analysis tools, such as Benford's Law and consistency analysis, as an effective means of detecting profits quality;

-The study recommends increased global attention to the anime motion picture sector, given its prominent role in supporting creative economy;

The following prospects for this study were identified:

-Future researchers may apply the methodology of Benford's law in other industries to compare and test the extent of profit integrity;

-Subsequent research could examine longer periods, or make comparisons across different countries, to see whether cultural, regulatory, or market conditions influence the integrity of reported profits;

Finally, this study provides practical evidence on the potential use of modern statistical methods to detect profit integrity in complex sectors, such as the motion picture industry, with the ultimate goal of ensuring fairness in financial reporting. Moreover, continuing to align accounting integrity with professional conscience helps strengthen investor and stakeholder confidence, reinforce the global competitiveness of film companies, and position them as a model for economic progress.

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Appendices

Table A1: The total study observations of Toei company during the period 2014-2024 (in Million ¥)

Year	Sales of Films	Licensing	Sales of Goods	Others Sales	Net Sales	SG&A Expenses	Gross Profit	Operating Income	Ordinary Income	Net Profit	Current Assets	Fixed Assets	Cash &Time Deposits	Current Liabilities	Fixed Liabilities	Shareholders' Equity	Net-to-Gross Profit Ratio (%)
2014 (FY15.3)	14155	10250	4628	1437	30313	5576	9572	3996	3978	2437	25914	24546	18144	7859	1497	39336	25.46
2015 (FY16.3)	14005	13803	4654	1320	33612	5167	12803	7635	7995	5145	31595	24808	19624	9499	1536	43868	40.18
2016 (FY17.3)	15939	18192	5531	1315	40747	5409	15543	10133	10362	7203	38909	27069	25591	12493	1936	49775	46.34
2017 (FY18.3)	16223	24124	4766	1068	45992	6309	17581	11272	11561	7847	40709	35050	28493	15744	1980	55849	44.64
2018 (FY19.3)	19531	30210	5166	970	55701	7097	22838	15741	16265	11375	55015	33476	34454	18871	2416	65246	49.79
2019 (FY20.3)	19925	29751	4401	911	54819	7367	23461	16094	16455	11437	60081	33937	39984	16701	2624	73669	48.73
2020 (FY21.3)	19766	28997	2466	446	51595	7436	22939	15503	16040	11067	64834	40643	43041	17623	2733	81894	48.26
2021 (FY22.3)	20769	32995	2231	1104	57020	8585	26693	18822	18822	12820	92929	33953	60149	29966	871	91849	48.04
2022 (FY23.3)	37267	42060	6149	2216	87457	12296	40966	28669	29791	20900	110702	39806	66909	35150	855	108768	51.04
2023 (FY24.3)	34828	39671	10681	3922	88654	14085	37449	23364	26453	18795	120455	42283	79007	28730	2295	121273	50.17
2024 (FY25.3)	37323	50582	9211	4315	100836	15989	48422	32432	33,188	23623	127940	63039	82474	34035	3745	138553	48.79

Note: End of Fiscal year of Toei Company is March 31

Source: Excel outputs based on financial statements retrieved from the official website of Toei Animation Company.